

BetterSearch

Semantic content search: retrieval by **meaning** rather than by shared words.

Search a content library for `Norman conquest` and get back articles about *1066*, the *Battle of Hastings* and the *Domesday Book* — none of which contain the phrase you typed.

Here is the same idea in a library catalogue. One reader, one phrasing, the same 74 records, and the only difference is which button is pressed.

Catalogue search — 0 results.

 Northfield Libraries

[My account](#) [Events](#) [Help](#)

RS

 R. Sandiford

Full catalogue ▾

Search

Catalogue search

Search by meaning

Try: [learning to be present](#) [gentle crime novels, nothing too gory](#) [anand](#) [getting my toddler to eat vegetables](#) [books to help me sleep better](#)

Home > Search results

Anywhere: learning to be present

Matched by **keywords** — records containing the words you typed.

Filter results

No results

< Previous Next >

Nothing in the catalogue matched those words.

The catalogue holds 74 items. A keyword search can only return a record that literally contains what you typed — if a reader's wording differs from the catalogue's, the item cannot be found at all.

Switch to **Search by meaning** to see what is actually here.

Search by meaning — 8 results.

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Full catalogue Search

Catalogue search Search by meaning

Try: [learning to be present](#) [gentle crime novels, nothing too gory](#) [anand](#) [getting my toddler to eat vegetables](#) [books to help me sleep better](#)

Home > Search results

Anywhere: learning to be present

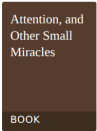
Matched by **meaning** — records about this subject, whatever words they use.

Filter results Showing 1 – 8 of 8 < Previous Page 1 Next >

Availability ▾
☐ Available now 8
☐ Available online 1

Format ▾
☐ Book 6
☐ Large print 1
☐ eAudiobook 1

Fiction / Non-fiction ▾
☐ Non-fiction 8



BOOK

Attention, and Other Small Miracles
Verhoeven, Sanne
Bellwether, 2017
No summary in this record.
Contemplation Psychology
Place reservation Available: 2 · Copies: 4 · Fenwick Road



Living Lightly

Living Lightly

Not one of those eight records contains the word *learning* or the word *present*. Six of the eight have no summary at all — nothing but a title, an author and a subject heading.

Why this exists

Keyword search (BM25, Postgres full-text, Elasticsearch's default) ranks by term overlap. If the user's words are not in the document, the document cannot be found — however obviously relevant it is. That is the entire failure mode this proof of concept addresses.

Semantic search embeds text as vectors positioned by meaning, so "Norman conquest" lands near "the invading army of Duke William" without sharing a single content word.

Install and run

```
pip install -e ".[dev,local,api]" # local backend: no API key needed
bettersearch ingest --corpus data/corpus_seed.json
bettersearch compare "Norman conquest" # the demonstration
bettersearch evaluate --per-query # the measurement
```

Then for the side-by-side UI:

```
uvicorn api.main:app --reload # http://127.0.0.1:8000
```

Nothing above needs an API key, a database, or network access after the first model download. A fresh clone has **no index** — `.bettersearch/` is gitignored, so `ingest` must run before any search.

For deployment, expected outputs and troubleshooting, see [RUNBOOK.md](#).

The catalogue prototype

The screenshots above come from a working page, not a mockup. It is the same library, the same index and the same `Searcher` — a catalogue-shaped front end over it, for showing the idea to people who will never read an nDCG table.

```
export BETTERSEARCH_INDEX_PATH=.bettersearch/catalogue
bettersearch ingest --corpus data/catalogue_library.json
uvicorn api.main:app --reload      # http://127.0.0.1:8000/catalogue
```

Three queries are worth trying, in this order:

1. **learning to be present** — the pair above. Catalogue search cannot return a record that does not contain the words; meaning-based search returns eight.
2. **gentle crime novels, nothing too gory** — a request phrased the way a reader actually asks. Keyword search finds three records and gets the point exactly backwards: it leads with *Nine Grams*, an organised-crime thriller, and picks up a book on spiritual life called *Nothing to Do* because the record contains the word "nothing". Meaning-based search opens with *The Knitting Circle Murders* and *Tea, Cake and Arsenic*.
3. **anand** — the failure that is easiest to miss, because it looks like it worked:

Anywhere: anand

Matched by **keywords** — records containing the words you typed.

Filter results

Availability

☐ Available now 9

Format

☐ Book 9

☐ DVD 1

Fiction / Non-fiction

☐ Non-fiction 6

☐ Fiction 4

Publication date

☐ 2010s 3

☐ 2020s 3

☐ 1990s 1

☐ 2000s 1

Location

☐ Ashcombe Branch 3

☐ Fenwick Road 3

☐ Northfield Central 2

☐ Marsden Park 1

☐ Rowley Green 1

Showing 1 – 10 of 10

< Previous Page 1 Next >

1.

[Punjabi book]

[Punjabi book]

Anand, Prem, 1931-1990

UUUU

No summary in this record.

Spiritual life

Place reservation

Available: 2 Copies: 3 · Ashcombe Branch

2.

Nothing to Do

Nothing to Do

Anand, Prem, 1931-1990

Wayfarer Press, UUUU

No summary in this record.

Spiritual life

Place reservation

Available: 3 Copies: 5 · Fenwick Road

3.

The Knitting Circle Murders

The Knitting Circle Murders

Anand, Ravi

Pennyfarthing, 2021

No summary in this record.

Detective and mystery stories

A surname is not an identity. Two unrelated authors share one, so keyword search files a crime novel in among books on spiritual life and looks perfectly healthy doing it. Notice the first record as well: no real title, no summary, no publication date. That record is not a contrived example — a catalogue of any size is full of them, and there is nothing there for a keyword search to match.

The catalogue is a fictional service with invented records, so it can be shown around without passing as a real library. Rebuild it with `python scripts/build_catalogue.py`, and regenerate these screenshots with

`python scripts/capture_screenshots.py` against a running server.

The three modes

MODE	HOW IT RANKS	GOOD AT	BLIND TO
keyword	BM25 term overlap	exact names, dates, figures, jargon	anything phrased differently
semantic	cosine over embeddings	paraphrase, concepts, questions	precise tokens it blurs together
hybrid	reciprocal rank fusion of both	most real traffic	—

`hybrid` is the mode you would normally reach for — though on this corpus it measurably loses to pure `semantic`; see the results below. Fusion uses **rank**, not raw score: BM25 scores are unbounded and cosine scores sit in $[-1, 1]$, so blending the numbers directly needs normalisation constants that must be re-tuned whenever the corpus or model changes. Ranks are comparable by construction.

Measured results

Run `bettersearch evaluate` to reproduce. 112 documents, 32 hand-labelled queries, `all-MiniLM-L6-v2` embeddings, `numpy` index.

MODE	RECALL@5	MRR@10	NDCG@10
keyword	0.355	0.625	0.415
semantic	0.841	0.984	0.890
hybrid	0.624	0.826	0.741

Semantic retrieval more than doubles nDCG over keyword and wins **30 of the 32 queries** outright. On eight queries — including `Norman conquest`, `why does bread rise`, `father of genetics` and `vaccination` — keyword search scores exactly **0.000**: not a poor ranking, but no relevant document retrieved at any depth.

Two illustrative failures worth seeing in the UI:

- `Norman conquest` → BM25 returns **nothing**. No document contains either word.
- `why does bread rise` → BM25 returns **Inflation**, because that article says "a sustained *rise* in the general price level". Term overlap without meaning.

Hybrid loses here, and that is a real result

Fusing a strong retriever with a weak one produced a *worse* ranking than the strong one alone (0.741 vs 0.890). That is not a bug in the fusion — it is what RRF does when one input list is mostly empty or noisy, which is the case on an eval set deliberately weighted towards keyword-hostile phrasing.

The honest reading: **ship semantic as the default for this kind of content**, and revisit `hybrid` only against a query mix with more exact-name, code and figure lookups, where BM25 earns its half of the fusion. Sample queries where keyword did contribute: `storing energy in a chemical form` (keyword

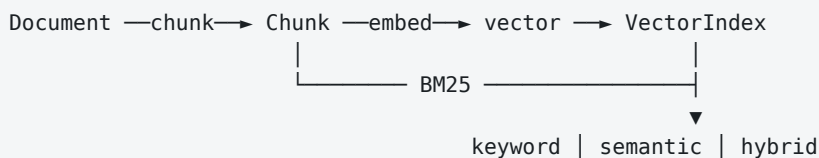
0.871 vs semantic 0.828) and 1066 (0.870 vs 0.876). Run `bettersearch evaluate --per-query` for the full breakdown.

Read this before quoting the numbers. The seed corpus and the eval labels were written by the same author, so the absolute values are indicative, not authoritative — the *gap between modes* is the signal. For a non-circular check, build a corpus you did not write:

```
python scripts/fetch_wikipedia.py --count 2000 --out data/corpus_wikipedia.json
bettersearch ingest --corpus data/corpus_wikipedia.json
bettersearch evaluate
```

Note also that the eval set is deliberately weighted towards keyword-hostile phrasing, because searching for words already in the text is the case keyword search already handles. Two control queries (Bayeux Tapestry, evolution by natural selection) use exact terminology, and BM25 is expected to win those.

Architecture



src/bettersearch/	
chunking.py	paragraph-aware splitting, overlap, content hashing
embeddings/	EmbeddingProvider protocol + local / OpenAI / Voyage
index/	VectorIndex protocol + numpy / pgvector
keyword.py	BM25 baseline
fusion.py	reciprocal rank fusion
search.py	Searcher – the public entry point
ingest.py	incremental ingestion
evaluate.py	Recall@5, MRR@10, nDCG@10
enrichment/	LLM enrichment of thin catalogue records
cli.py	
api/main.py	FastAPI wrapper - holds no retrieval logic
api/catalogue.py	catalogue-shaped presentation: facets, availability
web/index.html	developer comparison page
web/catalogue.html	the catalogue prototype

The library is framework-agnostic. `api/` and `cli.py` are thin callers, and neither is imported by the core.

`web/index.html` is the view for working on retrieval rather than for showing anyone — all three modes on one screen, scored, so a change in chunking or model is visible immediately:

BetterSearch

The same query, run three ways over the same 112 documents. Keyword search ranks by shared words; semantic search ranks by meaning. Try a query whose wording appears nowhere in the content.

Search query

Try: [Norman conquest](#) [why does bread rise](#) [the ship that sank on its maiden voyage](#) [father of genetics](#) [drugs that stop working](#)

Keyword search returned nothing for "Norman conquest". Semantic search found 5 results.

KEYWORD	SEMANTIC	HYBRID
BM25 term overlap. Cannot find a document that does not share your words.	Cosine similarity over embeddings. Matches meaning rather than vocabulary.	Reciprocal rank fusion of the other two columns.
<i>No results — no document shares these terms.</i>	0.428 The Battle of Hastings On 14 October 1066, an invading army under Duke William of Normandy met the English shield wall on Senlac Hill, a few miles inland from the coast of S...	0.016 The Battle of Hastings On 14 October 1066, an invading army under Duke William of Normandy met the English shield wall on Senlac Hill, a few miles inland from the coast of S...
	0.367 Harold Godwinson The last crowned Anglo-Saxon king of England, Harold was the most powerful nobleman in the kingdom before taking the throne in January 1066. His claim...	0.016 Harold Godwinson The last crowned Anglo-Saxon king of England, Harold was the most powerful nobleman in the kingdom before taking the throne in January 1066. His claim...
	0.328 Motte-and-Bailey Castles A raised earth mound topped by a wooden tower, with an enclosed courtyard below protected by a ditch and palisade. Cheap, fast to build, and defensibl...	0.016 Motte-and-Bailey Castles A raised earth mound topped by a wooden tower, with an enclosed courtyard below protected by a ditch and palisade. Cheap, fast to build, and defensibl...
	0.322 William, Duke of Normandy Born illegitimate in Falaise around 1028, William inherited his duchy as a child and spent two decades fighting to hold it. He claimed the English thr...	0.016 William, Duke of Normandy Born illegitimate in Falaise around 1028, William inherited his duchy as a child and spent two decades fighting to hold it. He claimed the English thr...
	0.316 The Battle of Britain Between July and October 1940 the Luftwaffe attempted to destroy British air defences as a precondition for invasion. Fighter Command was outnumbered ...	0.015 The Battle of Britain Between July and October 1940 the Luftwaffe attempted to destroy British air defences as a precondition for invasion. Fighter Command was outnumbered ...

112 chunks indexed · provider local · model sentence-transformers/all-MiniLM-L6-v2@384 · 384 dims · numpy backend

```
$ bettersearch compare "Norman conquest" # the same thing at the terminal
```

Configuration

Everything is a `BETTERSEARCH_*` environment variable; no code change is needed to switch provider or storage.

VARIABLE	DEFAULT	NOTES
BETTERSEARCH_EMBEDDING_PROVIDER	local	local openai voyage
BETTERSEARCH_INDEX	numpy	numpy pgvector
BETTERSEARCH_INDEX_PATH	.bettersearch/index	numpy backend only
BETTERSEARCH_DATABASE_URL	—	pgvector backend; falls back to DATABASE_URL
BETTERSEARCH_EMBEDDING_DIMENSIONS	512	Matryoshka width for API providers
BETTERSEARCH_CHUNK_TOKENS	450	target chunk size
BETTERSEARCH_CHUNK_OVERLAP	60	overlap between chunks
BETTERSEARCH_LOCAL_MODEL	sentence-transformers/all-MiniLM-L6-v2	any sentence-transformers model
BETTERSEARCH_LOCAL_TRUST_REMOTE_CODE	0	required by some long-context encoders; runs code from the model repo

API keys are read from `OPENAI_API_KEY` / `VOYAGE_API_KEY` / `ANTHROPIC_API_KEY` at the point of use and never stored by this package. Only enrichment needs an Anthropic key; indexing and search do not.

What actually breaks at scale

The embedding model is not the bottleneck people expect. Three other things are, and each has a specific countermeasure in the code.

1. Changing embedding model invalidates the whole index

Vectors from two models occupy different spaces. Comparing them returns confident nonsense — the worst kind of bug, because nothing errors.

Every vector is stored with its `model_id` and `dims`, and both index backends raise `ModelMismatchError` rather than answer a query embedded by a different model. That column is also what makes a *rolling* re-embed possible: without it, changing model means taking search offline.

2. Re-ingesting must not re-embed everything

Every chunk carries a `content_hash` over its embed text. Ingestion skips hashes already indexed, so a content update costs only the changed chunks. Re-run `bettersearch ingest` twice — the second run reports zero embedded.

3. Vector storage is the real wall

At 1M chunks × 1536 dims × 4 bytes you are holding **5.72 GB** of raw vectors before index overhead. Three multiplicative fixes, all implemented:

LEVER	EFFECT
Matryoshka truncation (1536 → 512)	3× smaller
halfvec fp16 storage	2× smaller
HNSW rather than flat scan	sub-linear query time

Together that is 6x: **5.72 GB** → **0.95 GB**, the difference between needing a large managed Postgres instance and fitting comfortably on a small one. Note the self-hosted MiniLM default is smaller still — 384 dims at fp16 is 0.72 GB for 1M chunks — so staying local sidesteps most of this rather than creating it.

4. Scaling while staying self-hosted

Self-hosted embeddings are the default path here: no per-token cost, no vendor dependency, and no content leaving your infrastructure — which matters more than the money if the corpus is ever commercially or personally sensitive.

Measure before you optimise. On this content the much-discussed 256-token cap costs exactly nothing:

```
$ # chunk length as all-MiniLM-L6-v2 itself tokenises it
  seed corpus (112 chunks)    min 75  median 87  p90 98  max 115
  catalogue   (74 chunks)    min 17  median 29  p90 49  max 57
  chunks exceeding the 256 cap: 0 (0%)
```

Measured with the model's own tokeniser, not the chunker's word-based estimate — the cap is the model's, so the model's count is the one that decides. Both corpora are short enough that no chunk reaches even half the cap, let alone the 450-token chunk target, so MiniLM truncates nothing. `bettersearch ingest` reports `truncated_chunks` precisely so this is a measurement rather than an assumption — check it against *your* content before changing anything.

The cap is architectural, not a compute limit. MiniLM stops at 256 tokens because its positional embeddings end there. More CPU or a bigger GPU makes it faster, never able to read more. If your chunks *do* exceed it, there are two fixes and only one of them costs anything:

1. **Shrink the chunks** — `BETTERSEARCH_CHUNK_TOKENS=200`. Free, no model change. Costs you more chunks (bigger index) and more fragmented context.
2. **Swap the model** — one env var, no code change. Costs compute and RAM.

```
export BETTERSEARCH_LOCAL_MODEL="BAAI/bge-base-en-v1.5"
bettersearch ingest --corpus data/corpus_seed.json # re-embeds under the new model
```

Dimensions and the input limit are read off the loaded model, so the index, the truncation warning and the `model_id` guard all follow automatically.

MODEL	INPUT	DIMS	SIZE	NOTES
all-MiniLM-L6-v2	256	384	80MB	current default
BAAI/bge-small-en-v1.5	512	384	130MB	2× window, same index size
BAAI/bge-base-en-v1.5	512	768	440MB	2× index size
intfloat/e5-large-v2	512	1024	1.3GB	
nomic-ai/nomic-embed-text-v1.5	8192	768	550MB	needs BETTERTSEARCH_LOCAL_TRUST_REMOTE_CODE=1
Alibaba-NLP/gte-large-en-v1.5	8192	1024	1.7GB	needs trust-remote-code
BAAI/bge-m3	8192	1024	2.3GB	multilingual

Bigger is not automatically better. Measured on this corpus, the 512-token model scored *worse* than MiniLM (nDCG 0.851 vs 0.890) — because nothing was being truncated, so the longer window bought nothing while the model itself happened to suit short passages less well. Re-run `bettersearch evaluate` after any model change; that is what the eval harness is for.

Measured CPU throughput on an ordinary container (no GPU), for a one-off backfill:

MODEL	CHUNKS/SEC (CPU)	1M CHUNKS
all-MiniLM-L6-v2	~119	~2.3 hours
bge-small-en-v1.5	~74	~3.8 hours

A single modern GPU is roughly 20–50× that, turning a backfill into minutes. Note this is a **batch** cost paid once per re-embed; query time is one embedding per search and is negligible either way.

The real steady-state cost of self-hosting is **RAM per web worker** (~600MB resident for MiniLM, more for larger models), not throughput. On a small Render instance that is the constraint that bites first — the usual fix is a separate embedding worker rather than loading the model into every web process.

If you ever do want a hosted provider

`openai` and `voyage` backends exist behind the same interface and are selected with `BETTERTSEARCH_EMBEDDING_PROVIDER`. **Neither has been exercised against a live API** — they are verified only as far as constructing, reporting correct dimensions, refusing to run without a key, and handling empty batches. Treat them as a starting point, not a supported path.

PROVIDER	1M CHUNKS (~400M TOKENS)	INPUT LIMIT
<code>openai</code> text-embedding-3-small	~\$8 (~\$4 batched)	8,191 tokens
<code>voyage</code> voyage-3.5-lite	~\$8	32,000 tokens

Voyage additionally supports asymmetric encoding (`input_type` of `document` vs `query`), which is a real recall gain and is wired in.

Enrichment for thin catalogue records

The production target is a library catalogue of 100k–1M items holding **thin records only** — title, author, subject headings. Around 15 words per item. That is not enough surface to embed: a catalogue record names a shelf position, while semantic search needs text that names *concepts*.

The enrichment pipeline generates that text with an LLM, and can state concepts the record never contains — turning `Subjects: Great Britain -- History` into prose mentioning the Norman Conquest, William the Conqueror and the Domesday survey. It builds the vocabulary bridge deliberately instead of hoping for it.

How much does thinness actually cost? (measured)

`scripts/synthesise_catalogue.py` strips the seed corpus down to thin records — title, synthetic author, LCSH-style subject headings, **no body text at all** — so the same 32 labelled queries can be re-run against a realistically impoverished corpus. The full-text score is the ceiling.

ARM	CORPUS	RECALL@5	NDCG@10
Ceiling	full article text	0.841	0.890
Baseline	thin records only	0.718	0.789
Treatment	enriched records	—	<i>needs an API key</i>

The aggregate drop is modest because these titles are unusually descriptive ("The Battle of Hastings" gives the game away). The per-query view is where the real damage shows — run `python scripts/compare_arms.py ceiling=... thin=...`:

QUERY	CEILING	THIN	DROP
planning a landing on a defended coast	0.871	0.296	−0.575
1066	0.876	0.369	−0.507
how electricity gets from a power station...	1.000	0.581	−0.419
evolution by natural selection	0.911	0.637	−0.274
Norman conquest	0.980	0.709	−0.271

The pattern is consistent and it is exactly the enrichment case: **queries collapse when the relevant documents' titles do not name the concept**. 1066 falls by more than half because *The Domesday Book*, *The Bayeux Tapestry*, *Stamford Bridge* and *Harold Godwinson* never contain the string. A synopsis would put it there.

Design

Three decisions carry the risk:

The generated text is a retrieval bridge and is never displayed. Match against it, then show the real catalogue record. A library is an authority-bearing institution; this one rule demotes a hallucination from a false statement by the library to a mildly bad search result.

Two modes, chosen per item by the model. Thin records maximise hallucination exposure — given only a title and a subject heading, the model draws on training knowledge rather than the record. So it first

reports whether it actually knows the work:

RECOGNISED	MODE	MAY USE
true	Knowledge synopsis	Its own knowledge of the work. Rich.
false	Grounded expansion	Only the record. Expand subject headings; invent nothing.

Grounded expansion is the safety floor and is still worth the call — turning authoritative controlled vocabulary into searchable prose is the bridge we want, with no fabrication risk. `bettersearch enrich` reports the split.

Enrichment is added alongside the record, never instead of it. The raw record stays indexed so the keyword lane can still match exact author, title and subject-heading terms. This is where `hybrid` should finally beat pure `semantic`, unlike on the article corpus above — real catalogue traffic is full of exact-name lookups, which is what BM25 is for. Re-measure rather than carrying the earlier conclusion across.

Prompt version is tracked like model id. A prompt change invalidates every stored enrichment, exactly as an embedding-model change invalidates every vector — except regenerating costs money rather than CPU. Every record stores `prompt_version`, `enrichment_model` and a `source_hash` of the input, so re-runs are incremental and a prompt edit is a costed, staged migration.

Running it

```
export ANTHROPIC_API_KEY=...
python scripts/synthesise_catalogue.py          # or bring your own catalogue
bettersearch enrich --corpus data/catalogue_thin.json --index
```

Uses the Batch API by default (half price, results keyed by `custom_id` since they arrive in arbitrary order). Resumable: everything already stored for this prompt/model/source is skipped, so a crash costs only the in-flight batch. `--sync` switches to one request per item for small runs.

Cost

Thin records invert the usual shape — input is ~100 tokens against a cached system prompt, output is ~300 tokens — so **output length dominates the bill**. Synopsis length is the primary lever, not model tier.

MODEL	1M ITEMS (BATCHED)	100K ITEMS
Haiku 4.5	~\$800	~\$80
Sonnet 5	~\$1,600	~\$160
Opus 5 (default)	~\$4,000	~\$400

At 100k the tier barely matters. At 1M it is a real decision — so measure it rather than guessing. Enrich a few hundred items at each tier into the same store (the model is part of the cache key, so tiers coexist) and compare with `scripts/compare_arms.py`. Embedding stays local and free either way.

Using the pgvector backend

```
docker compose up -d
export
BETTERSEARCH_DATABASE_URL="postgresql://bettersearch:bettersearch@localhost:5433/bettersearch"
export BETTERSEARCH_INDEX=pgvector
bettersearch ingest --corpus data/corpus_seed.json
bettersearch compare "Norman conquest"
```

Results should be identical to the numpy backend. The schema is created on first write and includes a generated `tsvector` column with a GIN index — the query to switch to when the corpus outgrows loading every chunk into memory.

HTTP API

ENDPOINT	PURPOSE
GET /health	provider, backend, chunk count, <code>model_id</code>
POST /search	{query, mode, top_k} → ranked results
POST /compare	one query, all three modes

Responses follow `{"status": "success", "data": {...}}`.

Dropping this into a Django/DRF project

The core library has no framework dependency, so the whole integration is one view:

```
# search/views.py
from bettersearch import Searcher
from rest_framework.response import Response
from rest_framework.views import APIView

_searcher = Searcher()          # module-level: loads the model once

class SemanticSearchView(APIView):
    def post(self, request):
        query = request.data.get("query", "").strip()
        if not query:
            return Response({"status": "error", "message": "query is required"}, status=400)
        response = _searcher.search(
            query,
            mode=request.data.get("mode", "hybrid"),
            top_k=int(request.data.get("top_k", 10)),
        )
        return Response({"status": "success", "data": response.to_dict()})
```

Point `BETTERSEARCH_DATABASE_URL` at the existing database and the chunk table lives alongside the rest of the schema. Embedding work belongs in a Celery task, not the request cycle.

Scope

Deliberately **not** included: LLM answer generation, query expansion, a reranker, authentication, and multi-tenant scoping. This PoC answers one question — does retrieval by meaning beat retrieval by keyword on this content — and each of those additions would obscure the answer.

Tests

```
pytest
```

Covers chunk boundaries and overlap, hash stability, the incremental-ingest skip, model-mismatch refusal, BM25 ranking, RRF ordering, numpy top-k against hand-computed values, and the full enrichment pipeline — resumability, prompt invalidation, per-model isolation, and out-of-order batch results. The tests use a deterministic fake embedding provider and a fake enrichment client, so they need no model, no network and no API key.